

RAGHAV KACHROO

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Software engineer drawn to friction points where systems are slow, manual, or breaking under load - building high-throughput data systems and ML inference infrastructure that fixes them at the root.

EDUCATION

University of California, San Diego Master of Science in Data Science (Artificial Intelligence & Machine Learning)	Sep 2024 – Mar 2026
Indraprastha Institute of Information Technology, Delhi Post Graduate Diploma in Data Science & Artificial Intelligence	Sep 2022 – Sep 2023
University of Delhi Bachelor of Management Studies	Jul 2018 – Jun 2021

EXPERIENCE

Hao AI Lab, UC San Diego Student Researcher Contributor to FastVideo — open-source video-generation inference framework; 3.7k+ stars, integrated as an NVIDIA Dynamo backend and adopted by SGLang .	Jan 2026 – Present San Diego, CA
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- Built a model-agnostic adaptive caching module for video-diffusion inference (residual-skip heuristic, no per-model adapters) delivering **22-47%** latency reduction on Wan2.1 across an SSIM **0.875-0.946** quality frontier.
- Surfaced a **23.7%** `torch.compile` speedup on Wan2.1-1.3B via Inductor fusion (**765** → **18** LayerNorm kernels).
- Registered `torch.library` custom ops with full `register_autograd` for FA2 attention, making attention traceable under `torch.compile` for both inference and training (**-13.5%** compile warmup on Wan2.1).
- Ported FastVideo to NVIDIA's DGX Spark (GB10/Blackwell): enabled **4-bit (FP4)** inference (**-24%** LTX2 denoise at 1080p) and shipped a sampling-preset fix (sharpness **92**→**431**) upstream.
- Cut Cosmos 2.5 inference latency **15.6%** on A100 by quantizing decoded frames on-GPU before device transfer.

Amazon Software Development Engineer Intern Built a distributed log indexing and query service over 42M+ log entries/hour using parallelized binary search, reducing incident triage latency from 15+ minutes to under 45 seconds . Automated on-call SOPs using AWS Step Functions and Lambda, saving 12+ engineer-hours per week . Exposed the log query service via MCP with caching and query batching, holding sub-2s response under load.	Jun 2025 – Sep 2025 Bellevue, WA
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Aark Global Software Developer, AI/ML Built an async document ingestion pipeline processing 18,000+ pages/day via Azure Queue Storage worker pool, maintaining throughput under variable load. Implemented read/write routing during datastore migration - Cosmos DB reads, MongoDB writes - maintaining sub-100ms P95 latency throughout cutover. Built a full-text search pipeline ingesting scanned PDFs through OCR into indexed Elasticsearch documents, enabling sub-180ms query latency over previously unsearchable content.	Apr 2023 – Sep 2024 Delhi, India
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PROJECTS & RESEARCH

Flare: ML-Powered Anomaly Detection Pipeline GitHub Built an end-to-end ML anomaly detection pipeline applying Drain3 template mining, Isolation Forest scoring, and DBSCAN clustering to 11.2M log lines across 575K blocks , achieving 0.642 F1 at ~40ms CPU-only latency.	Jan 2026 – June 2026
log-triage-v2: High-Throughput Log Indexing Engine GitHub Go log indexing engine with time-sorted in-memory index, concurrent write ingestion, and binary search reads - sustains 42M rows/hour with sub-millisecond nearest-timestamp lookup.	Jan 2026 – Apr 2026
OneBusAway / maglev: OSS Contributions GitHub Filed upstream fixes to the Go <code>routes-for-agency</code> endpoint (off-spec pagination, <code>includeReferences</code> handling) and merged a <code>StaleDetector</code> fix correcting missed staleness detection in GTFS-realtime feeds.	Jul 2026 – Present
TRAI: AI Mobile App to Reduce the Gap Between Triage and Care IEEE Publication	Jul 2025

TECHNICAL SKILLS

Languages: Python, Go, SQL, Java

Backend & Infra: FastAPI, REST APIs, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, MCP

Distributed & Data Systems: Kafka, Spark, Airflow, Elasticsearch, MongoDB, Cosmos DB, Azure Queue Storage, Redis

Observability & Cloud: Prometheus, OpenTelemetry, AWS (Lambda, Step Functions, S3), Azure

AI & ML: PyTorch, Scikit-Learn, Hugging Face Transformers, LangChain, MLflow, RAG Pipelines, Diffusers, LoRA

Concepts: Distributed Systems, Concurrency, MLOps, Inference Optimization, System Design, Model Monitoring